

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario-Based Assessment

DURATION: 1.5 Hrs

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario.

(5 Marks)

1. Introduction

A mobile app company wants to improve **user engagement** by sending personalized notifications at the most appropriate time. Instead of using fixed rules such as sending notifications at a particular time every day, the company can use **Reinforcement Learning (RL)** to learn the behavior of each user from their interactions with the app.

In this system, the RL agent observes the user's behavior, selects a notification-related action, receives a reward based on the user's response, and continuously improves its decision-making policy.

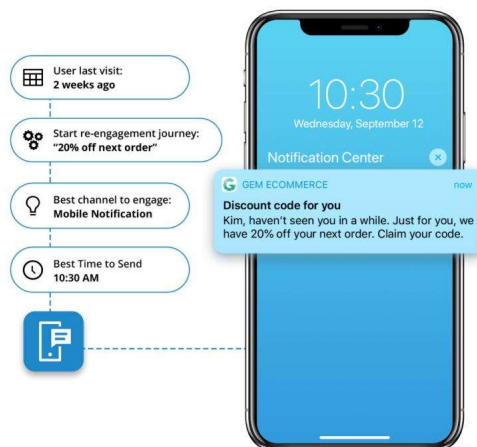
2. RL Framework

The RL framework consists of:

Agent → State → Action → Environment → Reward → New State

P. Sahith Sai
192425029
BTech AIML
Here:

- **Agent:** Personalized notification system
- **Environment:** Mobile app and user
- **State:** Current user behavior and context
- **Action:** Notification decision
- **Reward:** User engagement response
- **Policy:** Strategy used by the agent to select the best action



3. States

A **state** represents the current condition of the user. The notification system collects information about the user's recent behavior and uses it to determine the current state.

Possible state parameters include:

State Parameter	Example
Time of day	8 AM, 1 PM, 7 PM
Day	Weekday / Weekend
App usage	High / Medium / Low
Recent activity	Active / Inactive
Previous notification response	Opened / Ignored
Notification frequency	Low / Medium / High
User preference	Sports / Shopping / News
Session status	Active / Inactive

For example:

State S1:

Evening + User Active + Frequently Opens App + Previous Notifications Opened

This indicates that the user is likely to interact with a notification.

4. Actions

The **action** represents the decision taken by the RL agent.

Possible actions are:

1. **Send notification immediately**
2. **Delay notification**
3. **Send notification at a later time**
4. **Change notification content**
5. **Send a personalized notification**
6. **Do not send notification**

For example, if the user is currently inactive, the agent may select:

Action = Delay notification

If the user is active and historically responds positively in the evening:

Action = Send notification now

5. Reward Function

The reward function is one of the most important components because it tells the RL agent whether its decision was successful.

A suitable reward system can be:

User Response	Reward
User opens notification	+10
User clicks notification	+15
User performs desired action	+20
User reads notification	+5
User ignores notification	-5
User dismisses notification	-10
User disables notifications	-20

A simple reward function can be represented as:

$$R = \begin{cases} +20 & \text{if user performs desired action} \\ +10 & \text{if notification is opened} \\ +5 & \text{if notification is viewed} \\ -5 & \text{if notification is ignored} \\ -10 & \text{if notification is dismissed} \\ -20 & \text{if notifications are disabled} \end{cases}$$

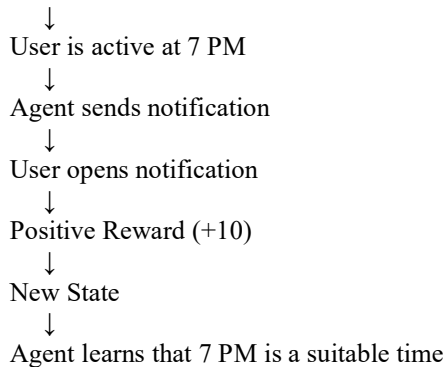
The system therefore learns to maximize **long-term user engagement** rather than simply maximizing the number of notifications sent.

6. Transition of States

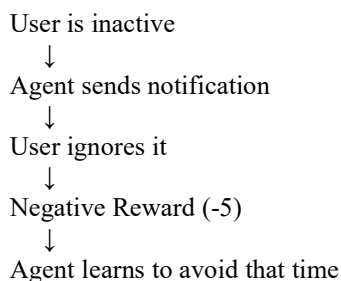
After an action is performed, the user's behavior may change, resulting in a **new state**.

For example:

Initial State



Similarly:



7. Policy

A **policy** determines which action the agent should take for a particular state.

Initially, the agent may not know the user's preferred notification time. It can therefore experiment with different times and actions.

After observing user responses, the agent gradually learns a better policy.

For example:

Time	User Response	Learned Preference
9 AM	Ignored	Low
1 PM	Viewed	Medium
5 PM	Clicked	High
7 PM	Opened and interacted	Very High

The agent eventually learns:

Best notification time $\approx$ 7 PM

8. Exploration and Exploitation

The system needs to balance **exploration and exploitation**.

The agent tries different notification times and formats to discover what works best.

Example:

- Try 6 PM
- Try 7 PM
- Try 8 PM

Exploitation

Once the agent learns that 7 PM produces high engagement, it sends more notifications around 7 PM.

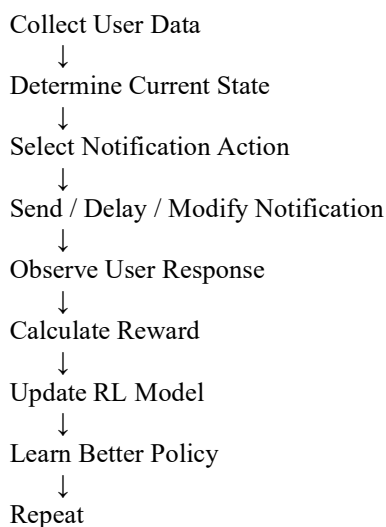
A common approach is the **ϵ -greedy strategy**:

- With probability ϵ , explore a new action.
- With probability $1 - \epsilon$, choose the action with the highest expected reward.

This prevents the system from becoming fixed to one behavior too early.

9. Learning Process

The complete learning process can be represented as:



The process continues continuously as the user interacts with the application.

10. Example Scenario

Consider a shopping application.

A user frequently opens the application between **7 PM and 9 PM**.

Initially, the RL system does not know this preference.

Day 1

The agent sends a notification at 10 AM.

User response: Ignored

Reward: -5

The agent learns that 10 AM may not be an effective time.

Day 2

The agent sends a notification at 5 PM.

User response: Notification opened.

Reward: +10

The agent learns that 5 PM is better.

Day 3

The agent sends a notification at 7 PM.

User response: User opens the notification and purchases a product.

Reward: +20

The agent learns that 7 PM is highly effective.

After repeated interactions, the agent learns that sending personalized notifications around **7 PM** gives higher engagement.

11. State-Action-Reward Example

A simple example can be represented as:

State	Action	User Response	Reward
Morning + inactive	Send	Ignored	-5
Afternoon + active	Send	Viewed	+5
Evening + active	Send	Opened	+10
Evening + active	Personalized offer	Purchased	+20
Night + inactive	Send	Dismissed	-10

The agent uses these experiences to improve its future decisions.

12. Q-Learning Approach

The company can implement **Q-Learning** to learn the best action for each state.

The Q-value represents the expected future reward of performing an action in a particular state.

The Q-learning update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where:

- ss = current state
- aa = selected action
- rr = received reward
- $s's'$ = next state
- $a'a'$ = next action
- α = learning rate
- γ = discount factor

The system updates the Q-value after observing the user's response.

13. Personalization

Different users may have different behavior patterns.

For example:

User A

- Most active: 7–9 PM
- Likes promotional notifications
- Frequently clicks notifications

User B

- Most active: 12–2 PM
- Prefers reminders
- Ignores promotional messages

The RL system can learn separate behavioral patterns and provide different notification strategies.

Therefore, instead of:

"Send the same notification to everyone at 7 PM."

The system learns:

"Send the right notification to the right user at the right time."

14. Advantages

The RL-based notification system provides several advantages:

1. **Personalization** – Notifications can be customized for individual users.
2. **Adaptive learning** – The system learns from changing user behavior.
3. **Improved engagement** – Notifications are delivered when users are more likely to respond.
4. **Reduced notification fatigue** – Unnecessary notifications can be avoided.
5. **Automatic optimization** – The system does not require manually defined rules.
6. **Long-term learning** – Previous interactions influence future decisions.
7. **Better conversion** – Relevant notifications can increase clicks, purchases, or app usage.

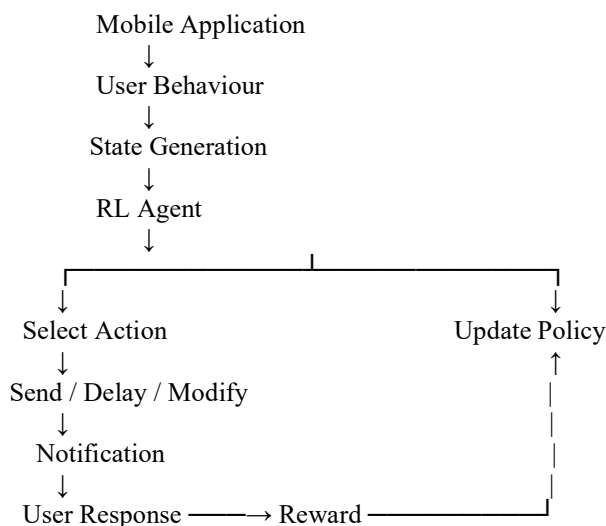
15. Challenges

Some challenges must also be considered:

- Large numbers of users and states
- Changing user preferences
- Sparse or delayed rewards
- Balancing exploration and exploitation
- Avoiding excessive notifications
- Privacy and protection of user behavioral data
- Cold-start problem for new users

For new users with little historical data, the system can initially use general user patterns and gradually personalize its decisions as more interactions are collected.

16. Overall Architecture



17. Conclusion

The mobile app company can model personalized notification delivery as a Reinforcement Learning problem. The user's behavior and context form the states, notification decisions form the actions, and user engagement provides the rewards. By continuously observing user responses and updating its policy, the RL agent learns the most effective notification timing and content for each user.

Thus, Reinforcement Learning can help the company maximize long-term user engagement while reducing irrelevant and unwanted notifications.

Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand